

PROJECT TITLE: RAINFALL, FOOD PRICES, AND EXTERNAL DEBT IN BANGLADESH - A PREDICTIVE AND CLUSTERING ANALYSIS

Course: MITM 431 - Big Data Analytics Project Proposal, Instructor: Dr. Shah Mostafa Khaled

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INTRODUCTION AND MOTIVATION

Bangladesh is one of the most climate-vulnerable countries in the world, where seasonal rainfall directly shapes agricultural output, food security, and macroeconomic stability. Yet existing research examines these variables in isolation. This project bridges that gap by constructing an integrated causal chain: Rainfall → Food Prices → External Debt. A growing body of literature supports each link individually. Auffhammer et al. (2012) showed that rainfall shocks significantly reduce rice yields across South Asia, with Bangladesh among the most affected. Ivanic and Martin (2008) demonstrated that food price spikes force developing governments to expand external borrowing to finance imports and social stabilization. Headey and Fan (2010) found that import-dependent nations like Bangladesh saw measurable increases in external debt during the 2007–2008 food crisis. Ahmed et al. (2020) confirmed that subnational rainfall anomalies within Bangladesh's divisional structure are strong predictors of crop output and market prices. Despite this evidence, no prior study has modeled the full pipeline within a single ML framework - this project fills that gap across 1999–2024.

RESEARCH QUESTIONS

- (1) **Does rainfall predict food prices?** - Can monthly rainfall levels and anomalies across eight divisions reliably predict staple food prices such as rice, flour, and oil? (Supervised Learning)
- (2) **Do food prices drive external debt?** - Can normalized food price trends predict Bangladesh's external debt indicators at the annual level? (Supervised Learning)
- (3) **Can years be grouped by overall risk level?** - Do natural clusters emerge when years are characterized by combined rainfall, food price, and debt profiles? (Unsupervised: K-Means)

DATA SOURCES AND PREPROCESSING PLAN

Three datasets covering 1999–2024 will be used. Sourced from Humanitarian Data Exchange by OCHA.

(1) **Subnational Rainfall Data (bgdrainfallsubnatfull.csv):** Filtered to adm_level = 1 across eight division PCodes - BD10 (Barishal), BD20 (Chattogram), BD30 (Dhaka), BD40 (Khulna), BD45 (Mymensingh), BD50 (Rajshahi), BD55 (Rangpur), BD60 (Sylhet). Retained columns: year, month, monthly_average_rf, anomaly, historic_rf, and pcode.

(2) **WFP Food Prices (wfp_food_prices_bgd.csv):** Filtered by division pcode, retaining high-availability commodities (rice, wheat/flour, cooking oil) as separate columns. All quantities normalized to per 1 KG or 1 L and converted to USD. Monthly averages computed per commodity per division.

(3) External Debt Indicators (externaldebt_bgd.csv): Exactly three key World Bank indicators selected, relevant to food import dependency and macroeconomic stress. Annual values repeated across all 12 months during merging.

The final merged dataset will have approximately 2,400+ rows ($26 \text{ years} \times 12 \text{ months} \times 8 \text{ divisions}$) with columns for year, month, pcode, rainfall features, normalized USD food prices, and three debt indicators.

TECHNOLOGY STACK AND PIPELINE

All three datasets will be loaded into Hadoop HDFS, with Hive used for SQL-style aggregation, joining, and monthly averaging. All machine learning will be performed in Python (Google Colab) using scikit-learn, pandas, numpy, and matplotlib.

METHODOLOGY

Supervised Machine Learning: Three algorithms will be applied to both prediction tasks (rainfall $\rightarrow$ food prices; food prices $\rightarrow$ external debt); Linear Regression as an interpretable baseline, Decision Tree following the reference script structure with `min_samples_split` tuned to prevent overfitting and the tree visualized as an image, and Random Forest as an ensemble extension to reduce variance. Multiple feature subsets will be tested across all models. 10-Fold Cross Validation will be applied throughout, with accuracy scores, confusion matrices, RMSE, and R-squared reported in clearly labeled tables and figures.

Unsupervised Machine Learning: K-Means Clustering will be applied to the full merged numeric feature space to group division-year combinations by overall risk profile. Optimal k will be determined via the Elbow Method and Silhouette Score, and each cluster interpreted qualitatively - for example, distinguishing high-rainfall, low-price, low-debt years from drought years with price spikes and elevated borrowing.

EXPECTED OUTCOMES AND CONTRIBUTION

The project will deliver:

- (1) A clean, merged dataset linking climate, food, and macroeconomic data for Bangladesh;
- (2) Cross-validated supervised ML models quantifying the rainfall-to-food-price, food-price-to-debt relationships;
- (3) Interpretable risk clusters across divisions and years;
- (4) A structured academic report with labeled figures, numbered axes, and written methodology explanations. Practically, if rainfall anomalies predict food price increases months ahead and food prices predict debt stress, then climate monitoring could serve as an upstream indicator for fiscal risk management in Bangladesh.

Approval